

**SIMATS ENGINEERING
ASSESSMENT TOOL I – MCQ**

Course: Data Handling and Visualization	Code: DSA0601	CO: CO1
Duration: 30 min	Total Marks: 25	Reg No : Name:

COURSE OUTCOME COVERED

CO1 Understand the fundamental concepts, principles, and techniques of data visualization. (BL2)

Activity Tool : MCQ

Weightage: 20%

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Q1. A hospital dashboard displays patient age using point size and blood pressure using point color. Elderly patients appear larger, while blood pressure is represented using a blue-to-red gradient. Which visualization principle is primarily violated?

- A. Position should not encode quantitative values.
- B. Point size is unsuitable for ordered quantitative comparisons.
- C. Color gradients cannot represent quantitative values.
- D. Age should always be mapped to the x-axis.

Q2. A financial analyst uses different shapes to represent monthly stock prices over time. Users struggle to identify trends. Which mapping would improve quantitative interpretation?

- A. Map stock prices to shape.
- B. Map stock prices to color hue.
- C. Map stock prices to vertical position.
- D. Map stock prices to transparency.

Q3. A researcher maps customer satisfaction levels (Very Poor, Poor, Fair, Good, Excellent) to random colors without ordering. The visualization becomes difficult to interpret because

- A. Satisfaction is nominal.
- B. Satisfaction is ordinal and requires ordered aesthetics.
- C. Random colors increase contrast.
- D. Color hue automatically implies ordering.

Q4. A climatologist maps rainfall intensity simultaneously to point size and color intensity. The visualization becomes misleading because

- A. Redundant encoding may exaggerate perceived differences.
- B. Color cannot represent quantitative variables.
- C. Point size cannot encode quantitative data.
- D. Multiple aesthetics always improve interpretation.

Q5. An analyst represents employee departments using a continuous color gradient. Which issue arises?

- A. Continuous scales imply ordering where none exists.
- B. Gradients improve categorical distinction.
- C. Departments are quantitative.
- D. Cartesian coordinates become invalid.

Q6. A scatter plot maps income to x-position and education level to y-position. Gender is represented using color. Which aesthetic represents nominal data?

- A. x-position
- B. y-position
- C. Color
- D. Position

Q7. A visualization maps annual sales to point transparency instead of position. Why is this a poor choice?

- A. Transparency has low perceptual accuracy.
- B. Transparency cannot encode variables.
- C. Transparency represents categories only.
- D. Position should only represent nominal data.

Q8. A chart uses texture patterns instead of colors to distinguish land-use categories. The primary advantage is

- A. Improved quantitative precision
- B. Better accessibility for color-blind users
- C. Reduced memory usage
- D. Faster rendering speed

Q9. A dataset contains Country, GDP, Population, and Continent. Which variable should ideally be encoded using shape?

- A. GDP
- B. Population
- C. Continent
- D. GDP Growth

Q10. Which visualization mapping produces the highest perceptual accuracy for comparing numerical values?

- A. Area
- B. Color saturation
- C. Position along a common scale
- D. Transparency

Q11. A data scientist uses a logarithmic scale for household income. Which reason best justifies this choice?

- A. Income contains many categories.
- **B. Income spans several orders of magnitude.**
- C. Log scales improve categorical comparison.
- D. Logarithmic scales remove outliers.

Q12. A temperature visualization assigns identical colors to values between 20°C and 25°C. The scale primarily reduces

- **A. Precision**
- B. Coordinate transformation
- C. Scatter density
- D. Position accuracy

Q13. A continuous variable is mapped to five discrete colors. Which problem is introduced?

- A. False precision
- **B. Information loss**
- C. Axis inversion
- D. Overplotting

Q14. A scatter plot uses identical symbol sizes for all observations while mapping sales to color intensity. Which aesthetic is unused?

- A. Position
- **B. Size**
- C. Color
- D. Shape

Q15. When transforming data values into visual positions, the mapping function is called

- A. Coordinate system
- **B. Scale**
- C. Axis
- D. Legend

Q16. A scientist rotates Cartesian coordinates by 45°. Which property remains unchanged?

- **A. Relative distances**
- B. Scale labels
- C. Color palette
- D. Shape encoding

Q17. A visualization requires comparison of radial distances from a central point. Which coordinate system is appropriate?

- A. Cartesian
- **B. Polar**
- C. Parallel
- D. Ternary

Q18. A business analyst compares quarterly revenues using pie slices arranged radially. Which issue is most likely?

- **A. Difficulty comparing angles accurately**
- B. Cartesian distortion
- C. Incorrect logarithmic transformation
- D. Missing legends

Q19. A map uses latitude and longitude. These coordinates are examples of

- A. Polar
- B. Cartesian
- **C. Geographic coordinate system**
- D. Logarithmic coordinates

Q20. Why are Cartesian coordinates preferred for scatter plots?

- A. They improve color perception.
- **B. They preserve independent quantitative dimensions.**
- C. They require fewer data points.
- D. They eliminate outliers.

Q21. A microbiologist visualizes bacterial counts ranging from 100 to 10 million. Which axis is most suitable?

- A. Linear
- B. Polar
- **C. Logarithmic**
- D. Circular

Q22. Which statement about logarithmic axes is TRUE?

- A. Equal distances represent equal differences.
- B. Equal distances represent equal ratios.
- C. Values cannot be negative.
- **D. Both B and C.**

Q23. A student uses a log axis for exam scores between 60 and 100. The visualization becomes

- A. More informative
- **B. Less meaningful because data span is small**
- C. Easier to compare
- D. Better for categorical data

Q24. Applying nonlinear scaling mainly helps

- A. Increase categorical distinction
- **B. Compress skewed numerical distributions**
- C. Improve color accessibility
- D. Eliminate missing values

Q25. Which transformation best linearizes exponential growth?

- A. Square root
- **B. Logarithm**
- C. Reciprocal
- D. Mean normalization

Q26. Which visualization naturally uses polar coordinates?

- A. Heatmap
- **B. Radar chart**
- C. Scatter plot
- D. Histogram

Q27. A wind-direction visualization requires angle and distance measurements. Which coordinate system is ideal?

- A. Cartesian
- **B. Polar**
- C. Geographic
- D. Matrix

Q28. Pie charts primarily encode data through

- A. Length
- B. Area
- **C. Angle**
- D. Transparency

Q29. Why are bar charts generally preferred over pie charts?

- **A. Bars compare lengths more accurately than angles.**
- B. Pie charts cannot display percentages.
- C. Bars require fewer colors.
- D. Pie charts cannot represent categorical data.

Q30. A circular timeline is chosen for seasonal events because

- **A. Seasons are cyclic.**
- B. Cartesian axes are unavailable.
- C. Polar coordinates increase precision.
- D. Colors require circular layouts.

Q31. A disease severity map uses green→yellow→red colors. This

represents · A. Qualitative scale

- **B. Sequential scale**
- C. Diverging scale
- D. Nominal scale

Q32. A map shows temperature anomalies centered at zero. Which palette is most suitable?

- A. Sequential
- **B. Diverging**
- C. Qualitative
- D. Random

Q33. Which color scale best distinguishes political parties?

- A. Sequential
- **B. Qualitative**
- C. Diverging
- D. Continuous grayscale

Q34. A visualization assigns identical colors to two unrelated categories. This primarily affects

- A. Accuracy
- **B. Distinguishability**
- C. Axis scaling
- D. Coordinate mapping

Q35. Which palette is inappropriate for categorical variables?

- A. Qualitative
- **B. Sequential**
- C. Distinct hues
- D. Random but unique colors

Q36. A transportation map uses different colors for bus, metro, train, and tram routes. Color serves to

- A. Encode quantitative values
- **B. Distinguish categories**
- C. Represent uncertainty
- D. Encode rankings

Q37. Why should adjacent categories use highly distinguishable colors?

- **A. Improve perceptual separation**
- B. Reduce axis labels
- C. Compress data
- D. Improve logarithmic scaling

Q38. A visualization uses red and green only. Which users may face difficulty?

- **A. Color-blind viewers**
- B. Statisticians
- C. Designers
- D. Programmers

Q39. A qualitative palette should primarily maximize

- A. Saturation ordering
- **B. Hue differences**
- C. Brightness ordering
- D. Transparency

Q40. When many categories exist, the best strategy is

- A. Use random colors.
- **B. Combine color with shapes.**
- C. Use grayscale only.
- D. Increase transparency.

Q41. A heatmap uses darker shades for higher values. This is an example of

- **A. Sequential color encoding**
- B. Qualitative mapping
- C. Shape encoding
- D. Nominal mapping

Q42. Which palette best emphasizes values above and below average?

- A. Qualitative
- B. Sequential
- **C. Diverging**
- D. Rainbow

Q43. Using rainbow palettes for ordered data may confuse viewers because

- **A. Hue changes lack perceptual ordering.**
- B. Rainbow colors are inaccessible.
- C. They reduce rendering speed.
- D. They eliminate legends.

Q44. A financial dashboard colors profits blue and losses orange around zero. This

is · A. Sequential mapping

- **B. Diverging mapping**
- C. Qualitative mapping
- D. Nominal mapping

Q45. Which aesthetic best complements color for quantitative heatmaps?

- A. Shape
- **B. Position**
- C. Size
- D. Transparency

Q46. A dashboard displays all bars gray except the current month in red. The red color is used to

- A. Encode categories
- **B. Highlight important information**
- C. Represent magnitude
- D. Encode uncertainty

Q47. Overusing highlight colors causes

- A. Improved focus
- **B. Reduced emphasis**
- C. Better quantitative comparison
- D. Improved accessibility

Q48. A researcher wants readers to notice only failed experiments. Which color strategy is best?

- A. Bright colors for all experiments
- **B. Neutral colors with failures highlighted**
- C. Sequential palette
- D. Random colors

Q49. A visualization will be printed in grayscale. Which palette choice is most appropriate?

- A. Random hues
- B. Equal luminance colors
- **C. High luminance contrast**
- D. Rainbow palette

Q50. A global health dashboard must be interpretable by users with various forms of color vision deficiency. Which design decision is best?

- A. Use only red and green.
- **B. Use color-blind-friendly palettes and redundant encodings.**
- C. Use maximum saturation for every category.
- D. Avoid legends.